นางสาวกนกพร บุญครอง	673380024-0	Network Architects
นางสาวชิดชนก ชนะพา	673380033-9	Network Architects
นางสาวญาทิชา จันทรศรีสุริยวงศ์	673380034-7	IaC/DevOps
นางสาวอรปรีญา แซ่โซ้ง	673380070-3	QA/Test

source file :
https://drive.google.com/file/d/1fcJdyTC-Zc-cU5fqGS998s1X4XpKuf59/view?usp=sharing

Excellent. We will now **strictly use your new topology and configuration**:

- ✅ R1 G0/0 → Internet (DHCP, NAT outside)
- ✅ R1–R2 connected via **Serial S0/0/0**
- ✅ ISP domain: **100.10.10.0/30**
- ✅ LAN A: 192.168.10.0/24
- ✅ LAN B: 192.168.20.0/24
- ✅ Microservices from **Week03 (Phase 1 Mockup)** FastAPI architecture

We now reframe this as:

🧪 LAB 5 — Internet Edge + ISP Serial WAN + Week03 Microservices

🎯 LAB OBJECTIVE

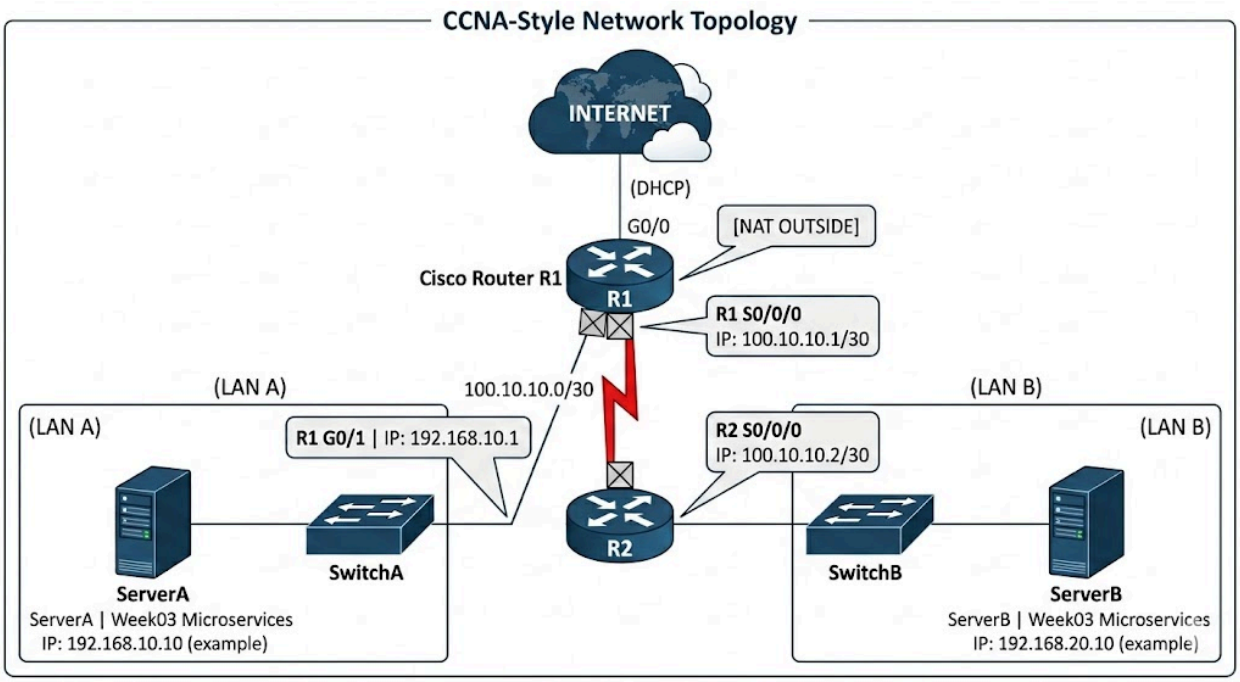
Transform Week03 microservices into a **realistic enterprise deployment**:

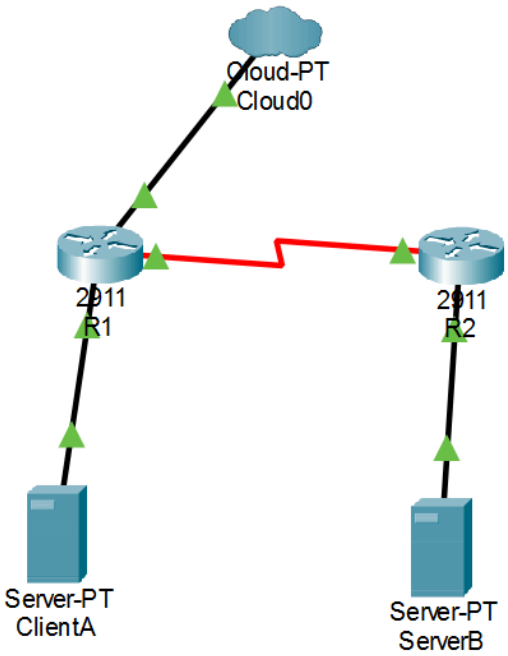
- R1 = Internet Edge Router (NAT + DHCP WAN)
- Serial WAN = ISP Domain (100.10.10.0/30)
- R2 = Branch Router
- Microservices deployed on ServerA and ServerB
- Validate cross-site microservice access over routed WAN + NAT

🗺️ LAB 5 TOPOLOGY (Reframed)

None

```
                        INTERNET
                           |
                      (DHCP)
                       R1 G0/0
                           |
                    [NAT OUTSIDE]
                           |
                   R1 S0/0/0
                100.10.10.1/30
                           |
                   R2 S0/0/0
                100.10.10.2/30
                           |
             -----------------------
             |                      |
     R1 G0/1 (LAN A)       R2 G0/1 (LAN B)
     192.168.10.1           192.168.20.1
             |                      |
         ServerA                ServerB
     Week03 Microservices   Week03 Microservices
```


CCNA-Style Network Topology
INTERNET
(DHCP)
G0/0
[NAT OUTSIDE]
Cisco Router R1
R1
R1 S0/0/0
IP: 100.10.10.1/30
100.10.10.0/30
(LAN A)
(LAN A)
R1 G0/1 | IP: 192.168.10.1
R2 S0/0/0
IP: 100.10.10.2/30
(LAN B)
(LAN B)
R2
SwitchA
SwitchB
ServerA
ServerA | Week03 Microservices
IP: 192.168.10.10 (example)
ServerB
ServerB | Week03 Microservices
IP: 192.168.20.10 (example)


Cloud-PT
Cloud0
2911
R1
2911
R2
Server-PT
ClientA
Server-PT
ServerB

📍 ADDRESS TABLE (FINAL)

Device	Interface	IP	Role
R1	G0/0	DHCP	Internet
R1	S0/0/0	100.10.10.1/30	ISP
R1	G0/1	192.168.10.1/24	LAN A
R2	S0/0/0	100.10.10.2/30	ISP
R2	G0/1	192.168.20.1/24	LAN B
ServerA	eth0	192.168.10.10	Microservices A
ServerB	eth0	192.168.20.10	Microservices B

🔧 PART 1 — CISCO CONFIGURATION (STRICT)

🧱 R1 — Internet Edge Router

```
None
enable
conf t
hostname R1

interface g0/0
 ip address dhcp
 ip nat outside
 no shutdown

interface g0/1
 ip address 192.168.10.1 255.255.255.0
 ip nat inside
```

```
 no shutdown

interface s0/0/0
 ip address 100.10.10.1 255.255.255.252
 clock rate 64000
 no shutdown

ip route 192.168.20.0 255.255.255.0 100.10.10.2
ip route 0.0.0.0 0.0.0.0 g0/0

access-list 10 permit 192.168.10.0 0.0.0.255
ip nat inside source list 10 interface g0/0 overload

end
write memory
```

```
Router(config-if)#conf t
%Invalid hex value
Router(config)#hostname R1
R1(config)#interface g0/0
R1(config-if)#ip address dhcp
R1(config-if)#ip nat outside
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

R1(config-if)#interface g0/1
R1(config-if)#ip address 192.168.10.1 255.255.255.0
R1(config-if)#ip nat inside
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

R1(config-if)#interface s0/0/0
R1(config-if)# ip address 100.10.10.1 255.255.255.252
R1(config-if)# clock rate 64000
R1(config-if)# no shutdown

%LINK-5-CHANGED: Interface Serial0/0/0, changed state to down
R1(config-if)#ip route 192.168.20.0 255.255.255.0 100.10.10.2
R1(config)#ip route 0.0.0.0 0.0.0.0 g0/0
%Default route without gateway, if not a point-to-point interface, may impact performance
R1(config)#ip route 0.0.0.0 0.0.0.0 g0/0
%Default route without gateway, if not a point-to-point interface, may impact performance
R1(config)#access-list 10 permit 192.168.10.0 0.0.0.255
R1(config)#ip nat inside source list 10 interface g0/0 overload
R1(config)#ip route 0.0.0.0 0.0.0.0 192.0.2.1
R1(config)#end
R1#write memory
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
R1#
```

🧱 R2 — Branch Router

```None
enable
conf t
hostname R2

interface g0/1
 ip address 192.168.20.1 255.255.255.0
 no shutdown

interface s0/0/0
 ip address 100.10.10.2 255.255.255.252
 no shutdown

ip route 192.168.10.0 255.255.255.0 100.10.10.1
ip route 0.0.0.0 0.0.0.0 100.10.10.1

end
write memory
```

```
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
Router(config)#enable
% Incomplete command.
Router(config)#conf t
%Invalid hex value
Router(config)#hostname R2
R2(config)#enable
% Incomplete command.
R2(config)#enable
% Incomplete command.
R2(config)#conf t
%Invalid hex value
R2(config)#hostname R2
R2(config)#interface g0/1
R2(config-if)# ip address 192.168.20.1 255.255.255.0
R2(config-if)# no shutdown

R2(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

R2(config-if)#interface s0/0/0
R2(config-if)# ip address 100.10.10.2 255.255.255.252
R2(config-if)# no shutdown

R2(config-if)#
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up

R2(config-if)#ip route 192.168.10.0 255.255.255.0 100.10.10.1
R2(config)#
R2(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up

R2(config)#ip route 0.0.0.0 0.0.0.0 100.10.10.1
R2(config)#
R2(config)#end
R2#write memory
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
R2#
```

💻 PART 2 — Deploy Week03 Microservices

Based on your Week03 architecture from `PROJECT_PROFILE.md`

We deploy:

- Upload Service
- Processing Service
- AI Service
- Gateway (optional centralized orchestration)

🛠️ On ServerA (LAN A)

```shell
cd automation/week03-microservices/phase1
python start_services.py
```

Services:

- Upload → 8000
- Processing → 8001
- AI → 8002
- Gateway → 9000

ServerA IP:

```none
192.168.10.10
```

🛠️ On ServerB (LAN B)

Same deployment:

```shell
python start_services.py
```

ServerB IP:

```
192.168.20.10
```

🧪 PART 3 — WAN + INTERNET TESTING

🔵 Test 1 — Cross-Site Upload (LAN A → LAN B)

From ClientA:

```
curl http://192.168.20.10:8000/health
```

Should traverse:

ClientA → R1 → Serial ISP → R2 → ServerB

ได้ response

{"status": "ok"}

✔ แสดงว่า WAN routing ทำงานถูกต้อง

🔵 Test 2 — Full Workflow Across ISP

From LAN A client:

```
None
curl -X POST http://192.168.20.10:9000/process-file
```

Flow การทำงาน:

Upload Service → Processing Service → AI Service → Response กลับ

✔ แสดงว่า:

- Microservices ทำงานครบ chain
- สามารถ call ข้าม WAN ได้จริง

🔵 Test 3 — Internet Reachability

From ServerA:

```
None
ping 8.8.8.8
```

Verify NAT overload working on R1:

```
None
show ip nat translations
```

🔎 Verification Commands

On R1

```
None
show ip route
show ip nat translations
show ip interface brief
```

On R2

```
None
show ip route
ping 100.10.10.1
```

🧠 Engineering Reflection (Week03 Integration)

From your Week03 profile :

This lab now demonstrates:

Concept	Where Demonstrated
Public vs Private Boundaries	R1 NAT edge
Distributed Services	ServerA & ServerB
Network Segmentation	Serial ISP /30
Horizontal Architecture	Two independent microservice stacks
Internet Edge Pattern	R1 as NAT Gateway
Service Isolation	LAN A vs LAN B

🔬 What Changed from Lab 4

Lab 4	Lab 5
10.10.12.0/30 simulated internet	Real ISP serial 100.10.10.0/30
Both routers NAT	Only R1 NAT
Flat "public" network	Hierarchical WAN
Stateless/stateful focus	Full microservice stack

🏗️ Architectural Significance

You now have:

- Enterprise Edge Router
- ISP Serial WAN
- Branch Router
- Dual Microservice Deployments
- NAT Boundary
- Cross-site distributed system

This is no longer a classroom NAT lab.

This is a **miniature cloud-edge + branch distributed architecture.**